Full User Manual: Laser Distance and Fill Level Sensor Modbus 4-20mA

Chapter 1. General Information and Principle of Operation

1.1 Device Purpose

The intelligent laser sensor is designed for continuous non-contact measurement of the distance to the surface of a monitored object, automatic calculation of the current fill level of bins, bunkers, tanks, or silos (in percent), and timely detection of emergency overfill thresholds.

Measurement data and alarm statuses are transmitted simultaneously via a digital industrial **RS-485 interface (Modbus RTU protocol)** and are formed as a continuous analog signal of an **active 4–20 mA current loop**. The device is designed to be integrated into process control systems (PCS), environmental monitoring complexes, and dispatch systems of industrial and warehouse facilities.

1.2 Time of Flight (ToF) Measurement Principle

The optical measurement module is based on the Time of Flight (ToF) method. The sensor periodically emits modulated light pulses in the near-infrared range (850 nm wavelength) and receives the signal reflected from the object. The computing core determines the time delay (phase shift) between the emitted and received waves, which is used to calculate the exact distance to the object.

The received physical distance data is fed into the built-in microcontroller, where it undergoes two-stage digital filtering:

1. **Moving median cascade:** Effectively cuts off random impulse noise and spikes caused by dust in the working area or strong structural vibrations.
2. **Exponential moving average (EMA) smoothing:** Stabilizes and smooths the readings, preventing phase jitter while maintaining a high response speed of the system to physical level changes.

Then the filtered distance is scaled according to the calibration coefficients, recalculated into the container fill level (0–100%), and matched with the discrete alarm setpoints.

1.3 Main Technical Specifications of the Device

- **Operating measurement range:** from 0.2 to 8 meters (depending on the target reflectivity).
- **Measurement resolution:** 1 mm (due to interpolation in the built-in filter).
- **Sensor polling frequency:** configurable, from 1 to 100 Hz.
- **Supply voltage:** 8…28 V direct current (VDC).
- **Power consumption:** not more than 3 W.

- **Communication interface:** RS-485, half-duplex (9600 bps, 8N1).
- **Analog output:** active 4–20 mA current loop (12-bit DAC resolution, load resistance of the loop — not more than 250 Ohm).
- **Non-volatile memory:** EEPROM (write endurance of at least 100,000 cycles).

Chapter 2. Installation Requirements and Optical Measurement Physics

To ensure stable and trouble-free operation of the laser rangefinder, the physical principles of ToF measurements of the semiconductor emitter must be strictly taken into account.

2.1 Blind Zone (Ranging Blind Zone)

The hardware blind zone of the sensor is **200 mm** (20 cm) from the front cover glass. Any objects closer than this distance cannot be correctly processed by the optical ToF receiver due to the photodiode saturation effect. If an object is closer than 200 mm, the sensor returns a value of `0 mm`. Programmatic or calibration elimination of the blind zone is not possible. The device must be mounted so that the maximum material fill level never comes closer than 220–250 mm to the sensor face (including a technological margin).

2.2 Target Reflectivity (Reflectivity) and Maximum Range

The maximum measurement range of the sensor directly depends on the diffuse reflection coefficient of the material's surface:

- **Light/white surfaces (90% reflection coefficient):** the maximum detection range is **8.0 meters**.
- **Dark/black surfaces (10% reflection coefficient):** the maximum range is reduced to **2.5 meters** due to infrared absorption by the material.

When monitoring the level of coal, coke, or wet dark soil, the effective measurement range narrows to 2.5 m. For monitoring the level of light-colored cement, lime, dry sand, or light plastic granules, the sensor operates stably at distances up to 8 m.

2.3 Field of View (FoV) and Light Spot Diameter

The laser beam of the emitter diverges in a cone with a full Field of View (**FoV**) of **2°** (half angle $\beta = 1°$). The measurement will be stable and accurate only if the light spot **completely** falls on the reflective surface of the measured material. The minimum required diameter of a flat target area d at a distance D is calculated by the formula:

$$d = 2 \times D \times \tan(1°) \approx 0.035 \times D$$

Light spot diameter vs distance table:

Distance to material (D), m	Minimum working spot diameter (d), cm
1,0	3.5
2,0	7.0
3,0	10.5
4,0	14.0
5,0	17.5
6,0	21.0
7,0	24.5
8,0	28.0

If the light spot partially hits two objects at different distances (e.g., the edge of the spot touches protruding rebar, a ladder, or the bin wall), the sensor will output an abnormal average distance with a high margin of error. The beam must be directed strictly perpendicular to the monitored plane and pass far from internal structures.

2.4 Influence of Ambient Light (Ambient Light Immunity)

The optical circuit of the sensor has built-in protection against ambient light up to **70 Klux**. Direct sunlight hitting the receiver lens can paralyze the receiver tract or drastically reduce accuracy. For outdoor mounting or installation in semi-open bins, the sensor must be protected by an opaque hood (visor). Directing the sensor at powerful artificial sources of infrared radiation (such as halogen floodlights) is not allowed.

2.5 Complex Surface Types and Obstacles in the Working Zone

- **Specular surfaces:** Highly reflective glossy surfaces (mirrors, polished metal, completely calm clean water) reflect the laser beam directionally away, leading it away from the sensor's receiver. To monitor the level of such media, the beam must be directed strictly at an angle of 90° to the plane, or a diffuse reflection target float must be placed on the specular surface.
- **Transparent media:** Ordinary glass, transparent acrylic, or water let the infrared beam pass through them. Mounting the sensor behind glass inspection windows is strictly prohibited.
- **Dustiness:** A dense suspension of dust at the moment of loading bulk material weakens the useful signal (the signal amplitude `Reg 1` drops sharply). In such conditions, it is necessary to use the maximum EMA damping level (`Reg 13` from 40% to 80%) to maintain stable readings during technological loading cycles.

Chapter 3. Electrical Connection

3.1 Cable and Power Requirements

The electrical installation of the device is performed using a round shielded cable (recommended type — KSPVE or shielded twisted pair). The cable shield is grounded on one side (on the control cabinet side) to minimize electromagnetic interference on the RS-485 line and the DAC. The supply voltage must be from 8 to 28 VDC. The internal sensor controller is hardware-protected against reverse polarity, but exceeding the supply voltage above 30 V will lead to irreversible failure of the device.

3.2 Connector Pinout and Color Marking

The connection is made through a sealed instrument connector. Below is the pinout layout on the soldering/cable connection side.

Pinout Assignment and Wire Colors:

Contact Number	Wire Color	Signal Name	Assignment
1	Red	`VCC`	Power supply plus (+8...28 VDC)
2	Blue	`GND`	Power supply minus / Common
3	Black	`RS485-A`	Direct line of the RS-485 half-duplex interface
4	Yellow	`RS485-B`	Inverse line of the RS-485 half-duplex interface
5	Green	`AO+`	Positive terminal of the active 4-20 mA current loop
6	White	`AO-`	Negative terminal of the active current loop (common ground)

3.3 Connection of the RS-485 Interface Line and the DAC Analog Output

1. **RS-485 Line:** At the ends of long network segments (over 150 meters), terminating resistors with a nominal value of 120 Ohm (0.25 W) must be installed between wires A and B. The network topology must strictly correspond to the 'bus' standard.

2. **4–20 mA Current Output:** The current generation circuit is **active** (the voltage for the current loop is formed inside the sensor from the supply voltage). The connection to an external PLC must be made to a **passive** analog input (measuring shunt). The total resistance of the PLC measuring shunt and the line wire resistance must not exceed **250 Ohm**.

Chapter 4. Modbus RTU Interface and Register Map

4.1 Protocol Parameters

Data exchange is carried out according to the Modbus RTU standard at a baud rate of 9600 bps, frame 8N1. The device acts as a Slave. Reading Holding Registers (`03h`) and Writing (`06h`, `10h`) functions are supported.

4.2 Detailed Register Map (Holding Registers)

Register Address	Parameter Name	Data Type	Access	Value Range	Parameter Description and Default Settings
0	Distance	`UInt16`	R	0…12000	Measured and filtered distance to the object (mm).
1	Sensor Signal	`UInt16`	R	0…65535	Amplitude of the reflected IR signal. Less than 100 — signal is weak.
2	Sensor Temperature	`Int16`	R	-40…125	Temperature inside the sensor enclosure (°C).
3	Calculated DAC Current	`UInt16`	R	400…2000	Current current loop value in hundredths of a mA (e.g., 1254 = 12.54 mA).
4	Fill Level	`UInt16`	R	0…100	Current container fill volume in percent (%).
5	Discrete Status	`UInt16`	R	0 / 1	Fill status: `0` — normal, `1` — threshold exceeded (overfill).
6	System Error Status	`UInt16`	R	0…3	Mask of active hardware device errors. Detailed in section 4.3.
7…9	Reserved Registers	`UInt16`	R	0	Used to align addresses of block reading. Return 0.

Register Address	Parameter Name	Data Type	Access	Value Range	Parameter Description and Default Settings
10	Modbus Network Address	`UInt16`	R/W	1…247	Device address on the network. Changes are applied immediately. *Default: 1*
11	Sensor Polling Frequency	`UInt16`	R/W	1…100	Frequency of trigger measurement pulses generation (Hz). *Default: 10*
12	Median Filter Window	`UInt16`	R/W	1…15	Sample size for the moving median. `1` — filter is off. *Default: 10*
13	EMA Damping	`UInt16`	R/W	0…100	Smoothing coefficient in %. `0` — filter is off. *Default: 20* (coeff. 0.8)
14	Calibration Point D1	`UInt16`	R/W	0…12000	Physical distance reference for the first point (mm). *Default: 500*
15	Calibration Point D2	`UInt16`	R/W	0…12000	Physical distance reference for the second point (mm). *Default: 5000*
16	Range Boundary X1	`UInt16`	R/W	0…12000	Distance (mm) corresponding to the minimum current (4 mA). *Default: 250*
17	Range Boundary X2	`UInt16`	R/W	0…12000	Distance (mm) corresponding to the maximum current (20 mA). *Default: 5000*
18	DAC Calibration Code Y1	`UInt16`	R/W	0…4095	12-bit DAC code for fine-tuning the 4.00 mA current. *Default: 819*
19	DAC Calibration Code Y2	`UInt16`	R/W	0…4095	12-bit DAC code for fine-tuning the 20.00 mA current. *Default: 4095*
20	Mode Control	`UInt16`	R/W	0…6	Service register for calibrations and tests. See section 4.4. *Default: 0*
21	Empty Bin Bottom D_empty	`UInt16`	R/W	0…12000	Distance to the bottom of the completely emptied container (mm). *Default: 5000*
22	Full Bin Top D_full	`UInt16`	R/W	0…12000	Distance to the 100% fill level (mm). *Default: 500*

Register Address	Parameter Name	Data Type	Access	Value Range	Parameter Description and Default Settings
23	DAC Output Mode	`UInt16`	R/W	0…2	Data source selection for DAC: `0` — distance, `1` — level, `2` — status.
24	«Full» Threshold	`UInt16`	R/W	10…100	Discrete status switching threshold in percent (%). *Default: 90*
25	Alarm Hysteresis	`UInt16`	R/W	1…50	Deadband zone for resetting the status in percent (%). *Default: 5*

4.3 Description of the System Error Register (Reg 6)

Register `Reg 6` is a bit mask of the current hardware status of the device. Reading this register by the upper level (PLC/SCADA) allows timely detection of an emergency situation on the object.

- **`0` (0x00) — System is healthy.** The ToF sensor transmits valid packets, the DAC successfully acknowledges transactions on the I²C bus.
- **`Bit 0` (0x01) — Rangefinder Communication Error.** This flag is set by the built-in software watchdog timer. If within three polling periods (3 * triggerInterval) no correct data packet is received from the measurement module, the system sets this bit. Once the data stream is restored, the error is automatically cleared.
- **`Bit 1` (0x02) — DAC Communication Error.** This flag is set if the microcontroller receives a NACK (No Acknowledge) signal from the DAC microchip when trying to write a new voltage value, or if the transaction was abnormally terminated by the hardware TWI timeout (3 ms).
- **`3` (0x03) — Multiple Failure.** Simultaneous failure of the optical module and the analog tract.

4.4 Description of the Mode Control Service Register (Reg 20)

Register `Reg 20` is designed to manage commissioning processes. Writing specific codes activates service scenarios:

- **`0` — Normal Mode.** The DAC is controlled according to the selected mode `Reg 23`.
- **`1` — Current Minimum Test.** The DAC forces the output voltage corresponding to code `Reg 18` (Y1). The output current must be equal to 4.00 mA.
- **`2` — Current Maximum Test.** The DAC forces the output voltage based on code `Reg 19` (Y2). The output current must be equal to 20.00 mA.
- **`3` — Capture Point M1.** The system reads the current distance and saves it to the EEPROM as the raw value of the first calibration point (M1).
- **`4` — Capture Point M2 and Calculation.** The system reads the current distance as the raw value of the second point (M2), calculates the calibration coefficients of the sensor, and automatically returns `Reg 20` to `0`.

- `5` — **Empty Bin Calibration.** The current filtered distance is captured and written to register `Reg 21` (D_empty). `Reg 20` is reset to `0`.
- `6` — **Full Bin Calibration.** The current distance is captured and written to register `Reg 22` (D_full). `Reg 20` is reset to `0`.

Chapter 5. Device Calibration Procedure

5.1 Two-point Sensor Calibration (D1/D2)

Calibration allows compensating for systematic error (zero offset and slope of the characteristic) caused by material reflective properties and the features of the device installation.

```
+-----------------------------------------------+
|  TWO-POINT SENSOR CALIBRATION (Reg 14-15)     |
+-----------------------------------------------+
                       |
                       v
+-----------------------------------------------+
|   Set reflector to reference D1 (mm)          |
|          Write to Reg 14 = D1                 |
+-----------------------------------------------+
                       |
                       v
+-----------------------------------------------+
|             Write to Reg 20 = 3               |
|        (Capture raw readings M1)              |
+-----------------------------------------------+
                       |
                       v
+-----------------------------------------------+
|   Set reflector to reference D2 (mm)          |
|          Write to Reg 15 = D2                 |
+-----------------------------------------------+
                       |
                       v
+-----------------------------------------------+
|             Write to Reg 20 = 4               |
|        (Capture M2, calculate & write)        |
+-----------------------------------------------+
```

1. Place a flat reflector in front of the sensor at a precisely measured distance (e.g., 500 mm). Write this value to `Reg 14` (`Reg 14 = 500`).
2. Send the first point capture command: write the value `3` to `Reg 20`.
3. Move the reflector to the second precisely measured distance (e.g., 4000 mm). Write it to `Reg 15` (`Reg 15 = 4000`).

4. Send the second point capture and calculation command: write the value `4` to `Reg 20`.
5. The microcontroller will automatically calculate the scaling factor and offset, write them to the EEPROM, and return `Reg 20` to `0`.

5.2 Container Calibration (Empty/Full Bin)

The procedure defines the physical boundaries of the tank to calculate the fill level in percent.

- **Step 1. Empty Bin Calibration:** Completely empty the container of material. Ensure that the laser beam falls on the bottom of the bin and does not hit foreign objects. Write the value `5` to `Reg 20`. The device will automatically save the current distance in `Reg 21` (D_empty) and switch to the operating mode.
- **Step 2. Full Bin Calibration:** Fill the container with material to the maximum allowable technological level. Write the value `6` to `Reg 20`. The sensor will measure the distance to the surface, save it to `Reg 22` (D_full), and return to the operating mode.

5.3 Built-in Protection Mechanism Against Invalid Parameters

The device's microcontroller continuously checks the parameters written by the user for physical and mathematical compatibility:

- **DAC Range Protection:** The `DistMin` (`Reg 16`) parameter must be strictly less than `DistMax` (`Reg 17`). The program maintains a minimum difference between them of at least **10 mm**. When trying to write `X1 >= X2`, the dependent parameter is automatically shifted by 10 mm forward or backward, preventing division by zero in the `map()` function.
- **Bin Range Protection:** The physical distance to the bottom of the empty bin `D_empty` (`Reg 21`) must be strictly greater than the distance to the full bin `D_full` (`Reg 22`). A minimum difference of **10 mm** is maintained.
- **Protection During Auto-capture (Modes 5 and 6):** If during the empty bin measurement the operator mistakenly writes a value when the material level is high (so that `D_empty` is less than the current `D_full`), the program automatically shifts the `D_full` value down, preserving the 10 mm delta and correcting the data in the EEPROM and Modbus registers. This ensures that the denominator in the level calculation formula never becomes zero and the system does not freeze.

Chapter 6. Analog Output Configuration and Calibration

6.1 Configuring the DAC Current Loop (4 and 20 mA)

To compensate for the tolerance of the resistors in the voltage-to-current converter circuit, perform the following steps:

1. Connect a milliammeter in series into the analog output circuit of the sensor (contacts `AO+` and `AO-`).

2. **Calibration of the 4.00 mA point:**
   - Write the value `1` to `Reg 20`. The output will be forced into the minimum current generation mode.
   - Modify the value of register `Reg 18` (Y1) up or down until the milliammeter reading is exactly **4.00 mA**.
   - Write `0` to `Reg 20` to save the calibration code Y1 in the EEPROM.
3. **Calibration of the 20.00 mA point:**
   - Write the value `2` to `Reg 20`. The output will switch to the maximum current generation mode.
   - Modify the value of register `Reg 19` (Y2) until the milliammeter reading is exactly **20.00 mA**.
   - Write `0` to `Reg 20` to save the calibration code Y2 in the EEPROM.

6.2 Description of Current Output Operating Modes (Reg 23)

The data source for forming the current signal is selected via register `Reg 23`:

- **`0` — 'Distance' Mode (default).** The output current is linearly scaled according to the distance between points `DistMin` (`Reg 16`) and `DistMax` (`Reg 17`).
  - If the distance narrows to `DistMin`, the current is 4 mA.
  - If the distance increases to `DistMax`, the current is 20 mA.
- **`1` — 'Level' Mode.** The output current is proportional to the container fill level in percent (0…100%).
  - At a fill level of 0% (distance is equal to or greater than `D_empty`), the current is 4 mA.
  - At a fill level of 100% (distance is equal to or less than `D_full`), the current is 20 mA.
- **`2` — 'Alarm' Mode.** The current output operates in discrete mode:
  - If `Reg 5` (fill status) is `0` (normal), the current is 4 mA.
  - If `Reg 5` is `1` (filled), the current is 20 mA.

Chapter 7. Diagnostics, Troubleshooting, and Factory Reset

7.1 Error Codes in Register 6 and Their Interpretation

When integrating the sensor into a SCADA system, it is recommended to configure continuous monitoring of the error register `Reg 6`:

- **Code `1` (Rangefinder Error):** Indicates no data from the laser rangefinder. It is possible that the internal ribbon cable of the sensor is damaged, no internal 5 V power is supplied, or the sensor lenses are covered with a thick layer of opaque dirt.
- **Code `2` (DAC Error):** Indicates a fault on the I²C bus. Possible causes include a physical break in the tracks to the DAC microchip, short-circuit of the SDA/SCL lines, lack of pull-up resistors, or failure of the DAC microchip due to a voltage spike in the current loop.

7.2 Typical Faults Table

Fault	Possible Cause	Remedy
No Modbus communication	1. No sensor power. 2. Swapped RS-485 A and B lines. 3. Device network address mismatch.	1. Verify with a multimeter the presence of 8...28 V voltage on contacts 1 and 2. 2. Swap wires on contacts 3 and 4. 3. Try polling address 1 or perform a network scan.
Distance readings are frozen at 0 mm	The sensor is in the blind zone (closer than 200 mm from the target).	Move the sensor further from the material surface, respecting the minimum working distance.
Distance fluctuates erratically	1. The material surface is unstable. 2. Foreign objects are inside the beam cone (2° FoV).	1. Increase the median window size (`Reg 12` to 12-15) and EMA damping (`Reg 13` to 30-50%). 2. Re-mount the sensor further from walls or ladders.
Fill level is always 0%	Container empty and full calibration was not performed, or the D_empty and D_full parameters are equal.	Perform calibration according to Chapter 5. Ensure that the empty container distance is strictly greater than the full container distance.
Current output delivers a fixed current (4 or 20 mA)	The sensor remained in the service test mode.	Write the value `0` to the control register `Reg 20` .

7.3 Factory Reset Procedure (Factory Reset)

To restore the device parameters to the factory default values, write the following values to the Holding registers:

- **Reg 10 (Network Address):** `1`
- **Reg 11 (Polling Frequency):** `10` Hz
- **Reg 12 (Median Window):** `10` samples
- **Reg 13 (EMA Damping):** `20` %
- **Reg 14 (Calibration Point D1):** `500` mm
- **Reg 15 (Calibration Point D2):** `5000` mm
- **Reg 16 (Point X1 for DAC):** `250` mm
- **Reg 17 (Point X2 for DAC):** `5000` mm
- **Reg 18 (DAC Code Y1):** `819` (corresponds to ~1 V)

- **Reg 19 (DAC Code Y2):** `4095` (corresponds to ~5 V)
- **Reg 21 (Empty Container D_empty):** `5000` mm
- **Reg 22 (Full Container D_full):** `500` mm
- **Reg 23 (DAC Mode):** `0` (distance mode)
- **Reg 24 (Threshold «Full»):** `90` %
- **Reg 25 (Hysteresis):** `5` %

Note: After writing these values, it is recommended to reboot the device's power to completely clear the RAM and initialize the filtering cascades.